Multi-Task

AID (Scene Classification)

***Q*:** Classify the image into following categories: [‘Airport’, ‘BareLand’, ‘BaseballField’, ‘Beach’, ‘Bridge’, ‘Center’, ‘Church’, ‘Commercial’, ‘DenseResidential‘, ⋯⋯, 'Square', 'Stadium', 'StorageTanks’, ‘Viaduct’]
***A*:** Airport

VRS-Cap (Image Captioning)

***Q*:** Could you describe the contents of this image for me?
***A*:** The high-resolution image from Google Earth shows a sports complex area with a soccer field with bright lines defining the playing areas. There are buildings adjacent to the fields, surrounded by trees. Several small vehicles are parked near the buildings and along the road at the bottom of the image.

VRS-VG (Visual Grounding)

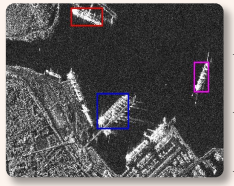
***Q*:** The tennis court is positioned on the left side of the image, characterized by its rectangular shape in a blue color.
***A*:** [440, 0, 910, 470]
***Q*:** The baseball diamond is identifiable in the upper right part of the image.
***A*:** [440, 0, 910, 470]

VRS-VQA (Visual Question Answering)

***Q*:** Are there any roads visible in the image?
***A*:** Yes
***Q*:** What color are the roofs of the tanks in the top-right and right-most positions?
***A*:** Light-colored

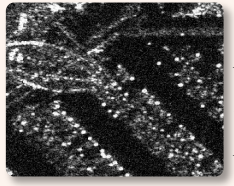
Multi-Modality

SAR-VG (Visual Grounding)



***Q*:** ship near the top edge of the image with a long, narrow shape.
***A*:** [300, 26, 446, 131]
***Q*:** ship on the far right side of the image with a tall, vertical structure.
***A*:** [862, 338, 930, 509]
***Q*:** ship in the lower center with a wide body and multiple visible sections.
***A*:** [420, 517, 566, 719]

SAR-VQA (Visual Question Answering)

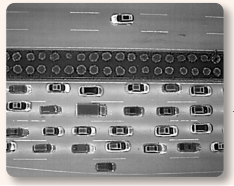


***Q*:** Are there any other categories apart from harbor and bridge?
***A*:** No
***Q*:** Where are most bridges positioned within the image? The classes are: top, center, bottom, left, right.
***A*:** Top

IR-VG (Visual Grounding)

***Q*:** The car located furthest from the crosswalk in the center of the image.
***A*:** [465, 326, 494, 413]
***Q*:** The car positioned closest to the right edge of the vehicle group near the crosswalk.
***A*:** [518, 496, 549, 576]
***Q*:** The car positioned closest to the crosswalk and the left side of the vehicle group.
***A*:** [406, 497, 439, 587]

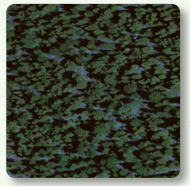
IR-VQA (Visual Question Answering)



***Q*:** What is the number of freight car in the image?
***A*:** 1

Multi-Application

FireRisk (Fire Risk Assessment)



***Q*:** Based on the remote sensing image, what is the fire risk level among the following choices? ['High', 'Low', 'Moderate', 'Non-burnable', 'Very_High', 'Very_Low', 'Water']
***A*:** Very_High

RescueNet (Hurricane Damage Assessment)

***Q*:** What degree of damage is there to the scene?
***A*:** Major Damage
***Q*:** Is there any building that has been minorly impacted by the disaster?
***A*:** Yes